Data Analytics I — Assignment 1 Report

Team ID: 30

Sale Sangameshwar 2024102017
Shaik Mastan Sharif 20241035

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1 Introduction

This report presents a comprehensive analysis of two datasets as part of the Data Analytics I Assignment:

1. **Bank Term-Deposit Marketing Dataset** (`bank.csv`): 11,162 records of direct marketing campaigns from a banking institution, with 17 features covering customer demographics, financial status, and campaign outcomes.
2. **Airbnb NYC Listings Dataset** (`listings.csv`): 30,259 short-term rental listings across New York City's five boroughs, with 90 features covering pricing, location, amenities, host information, and availability.

The analysis covers data cleaning, feature engineering, distribution analysis, group comparisons, statistical testing, and market segmentation. All results are computed from the actual datasets — no values are fabricated.

2 Part I — Bank Dataset

Data Cleaning Methods

Aspect	Finding	Decision
Missing values (NaN)	0 across all columns	No imputation needed
Duplicate rows	0	No deduplication needed
'Unknown' values	Present in `education` (497), `contact` (2,346), `poutcome` (8,326)	Retained; treated as 'Not-reported' where appropriate
Negative balances	688 records	Kept — valid overdraft amounts
`pdays = -1`	Sentinel encoding	Kept — indicates "not previously contacted"

Table 1: Bank dataset cleaning summary

The dataset is structurally clean with no missing values. The 'unknown' entries are coded non-responses, not system errors, and are handled contextually in each analysis section.

2.1 1. Education Distribution & Grouping

Observation: Four education levels exist: secondary (49.1%), tertiary (33.1%), primary (13.4%), and unknown (4.5%). Secondary education is the dominant category.

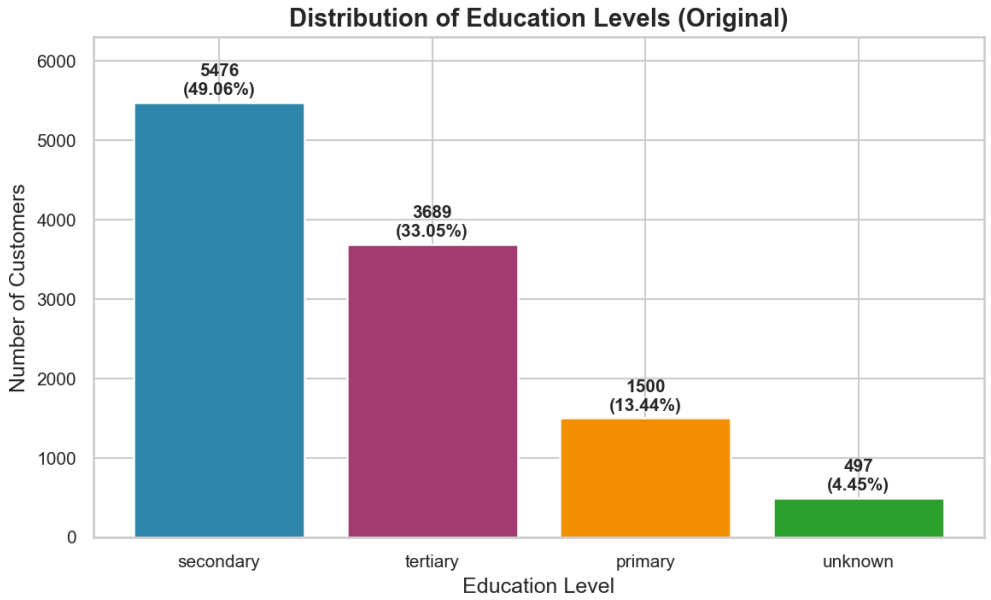


Figure 1: Original Education Levels Distribution

Grouping: A meaningful mapping was applied:

- `primary` → `Basic`
- `secondary` → `Intermediate`
- `tertiary` → `Higher`
- `unknown` → `Not-reported`

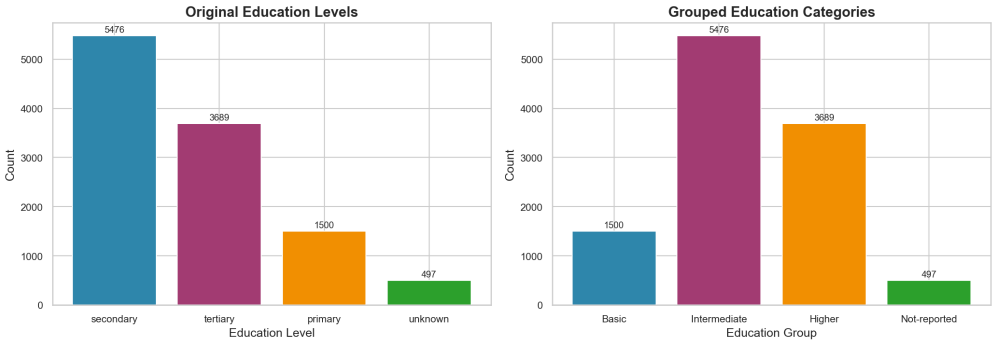


Figure 2: Grouped Education Categories Distribution

Interpretation: This renaming provides semantic clarity — 'Basic/Intermediate/Higher' are universally understood education tiers. Crucially, relabelling 'unknown' as 'Not-reported' prevents it from being treated as an actual education level, which would be misleading in any downstream analysis or model. As seen by comparing Figure 1 and Figure 2, since each original value maps 1:1 to a group, this is a semantic improvement, not an aggregation. The frequencies (counts and percentages) for each level remain exactly the same before and after the grouping. The value lies in clarity for stakeholders and consistent encoding for models.

2.2 2. Age–Contact Engagement Intensity Relationship and Grouping

Age Distribution

- **Range:** 18 to 95 years
- **Central tendency:** Mean = 41.2, Median = 39.0 (right-skewed)
- **Spread:** IQR = 32–49, covering the core working-age population
- **Shape:** Right-skewed with a long tail of elderly customers

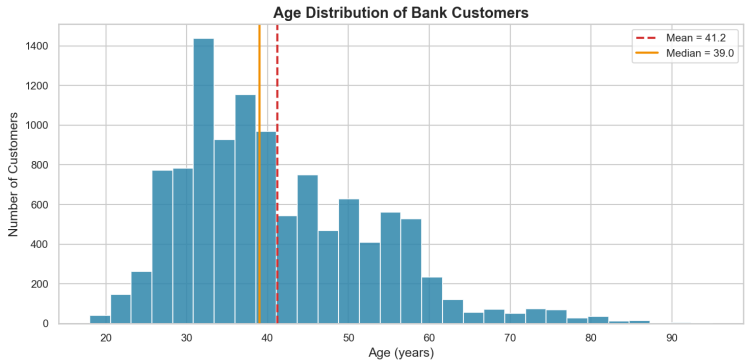


Figure 3: Age Distribution of Bank Customers

Call Duration Distribution (seconds)

- **Range:** 2 to 3,881 seconds
- **Central tendency:** Mean = 372.0s, Median = 255.0s (strongly right-skewed, skewness = 2.144)
- **Key insight:** The mean substantially overestimates the "typical" call duration because it is pulled upward by extreme long calls. The median (255s ≈ 4.3 min) is the more robust measure.

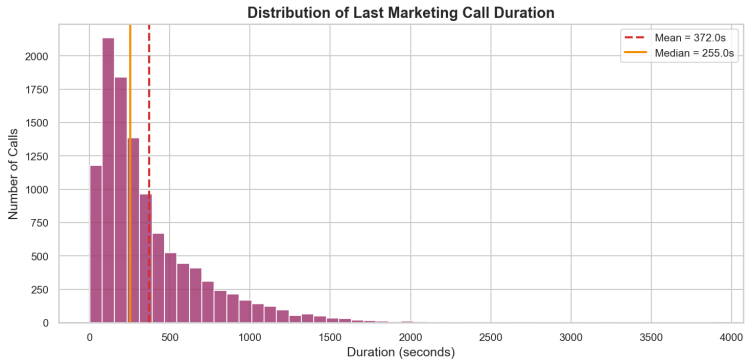


Figure 4: Last Contact Duration Distribution

Age Groups Created

Group	Range	Rationale
Young Adult	18–29	Entry-level, lower savings
Early Adult	30–39	Established career, family formation
Middle Adult	40–49	Peak earning years
Older Adult	50–59	Pre-retirement
Senior	60+	Retirement age

Table 2: Age group definitions

Contact-Intensity Groups (Quartile-Based)

Group	Duration Range	Basis
Brief	$\leq$ 138s	$\leq$ Q1
Standard	138–255s	Q1–Q2
Extended	255–496s	Q2–Q3
Deep-engagement	$>$ 496s	$>$ Q3

Table 3: Contact-intensity group definitions

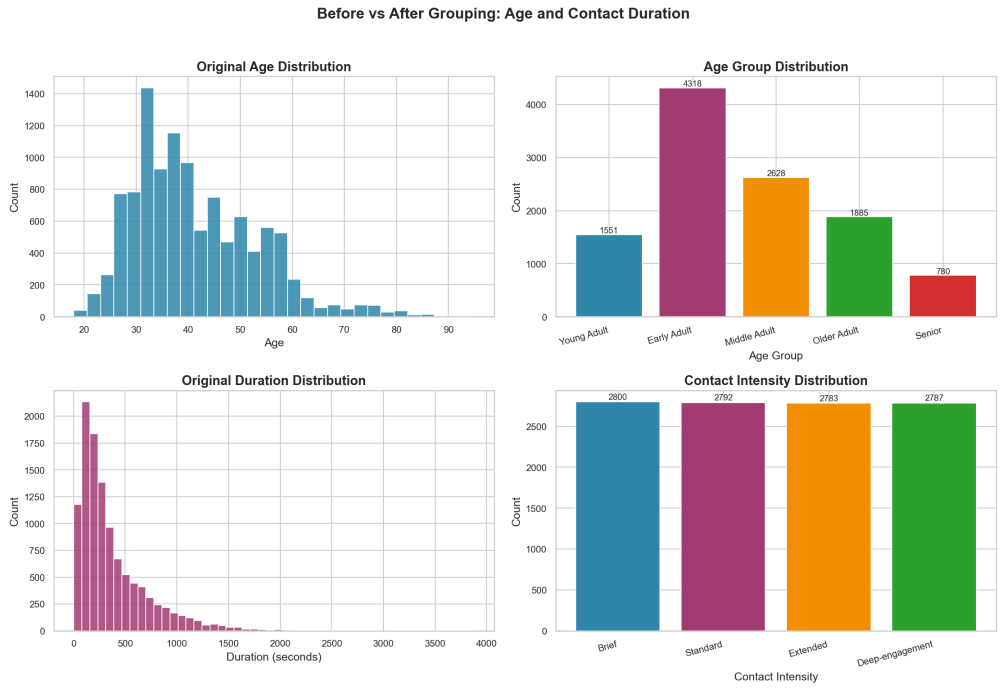


Figure 5: Distribution Before and After Grouping for Age and Contact Intensity

Relationship Analysis

Before grouping (numerical):

- Pearson $r = 0.0002$, $p = 0.9841$ — essentially zero linear correlation

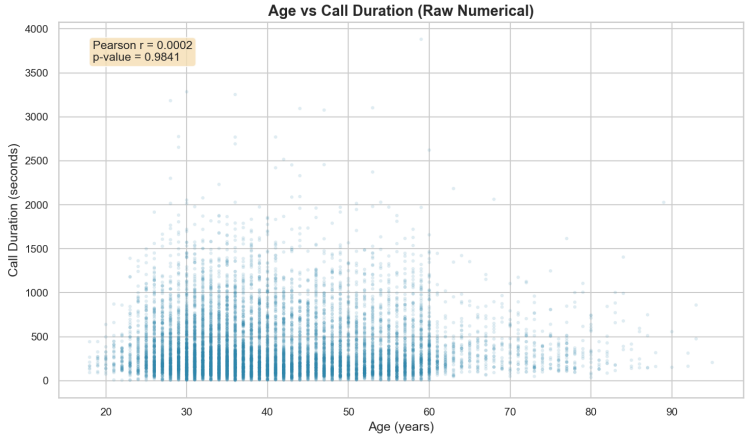


Figure 6: Scatter Plot of Age vs. Contact Duration (Original Numeric Values)

The scatter plot (Figure 6) confirms the lack of a discernible linear pattern between age and the duration of the contact.

After grouping (categorical):

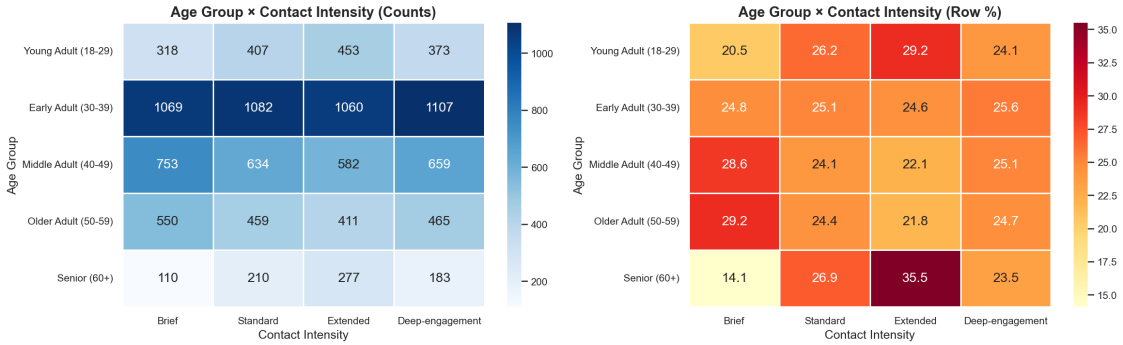


Figure 7: Heatmap of Age Group by Contact Intensity (Counts and Row Percentages)

The heatmap in Figure 7 highlights the value of looking at relative row percentages rather than absolute counts. While the raw count is dominated by the large "Early Adult" demographic, the row percentages show that Seniors (60+) actually have the highest concentration of "Extended" contacts (35.5%) compared to other age brackets.

- Chi-square statistic $= 144.14$, $df = 12$, $p < 0.0001$
- Chi-square test applied on the contingency table (age group × contact intensity)
- Cramér's V $= 0.066$ — indicates a negligible association despite statistical significance
- Consistent with the numerical analysis

Statistical evidence: The chi-square test and Cramér's V both confirm that age and call duration are essentially independent in this dataset.

Grouping evaluation: Grouping improves interpretability (categories are easier to communicate

and plot as heatmaps) but does not reveal any strong hidden relationship. The trade-off is justified because the lost numerical precision carried no analytical value.

2.3 3. Account Balance Composition and Group Performance

Overall Distribution

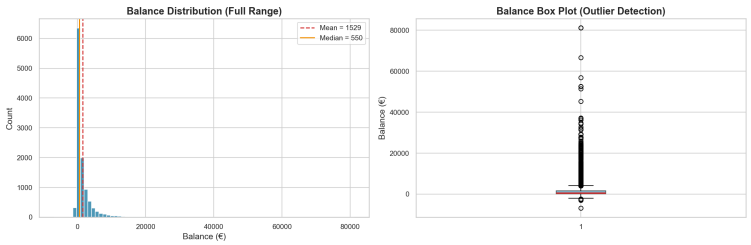


Figure 8: Overall Account Balance Distribution

- **Range:** −€6,847 to €81,204
- **Central tendency:** Mean = €1,529, Median = €550 (heavily right-skewed, as seen in Figure 8)
- **Outliers:** ∼13% of values fall outside the IQR fences

Balance Components

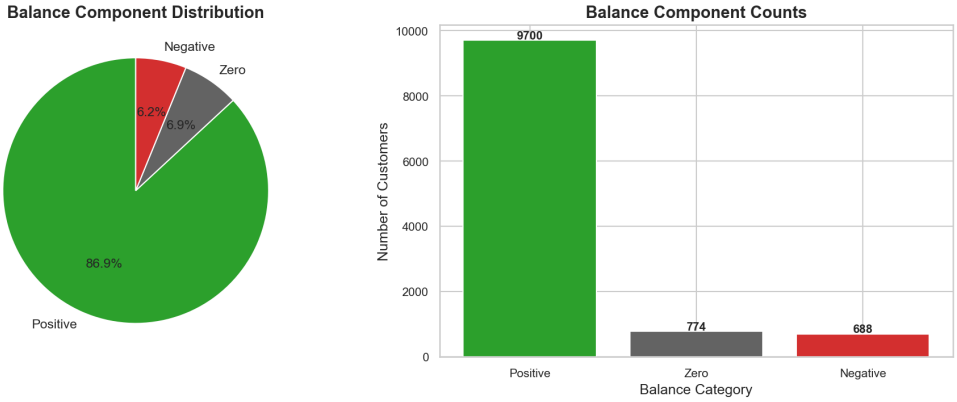


Figure 9: Balance Categories: Proportion and Average Values

Category	Count	Percentage	Avg Balance	Median Balance
Positive	9,700	86.9%	€1,782	€730
Zero	774	6.9%	€0	€0
Negative	688	6.2%	−€324	−€239

Table 4: Balance composition by category

The majority of customers (86.9%) have positive balances. Negative balances represent overdrafts and are modest in magnitude (Figure 9).

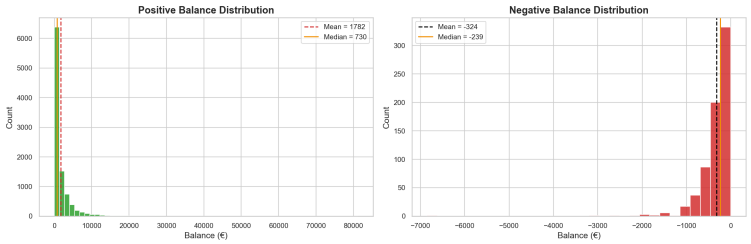


Figure 10: Distribution of Non-Zero Account Balances (Log Scale)

Looking at non-zero balances on a logarithmic scale (Figure 10), the wide spread and long right tail of positive balances becomes more visually apparent.

Group Performance

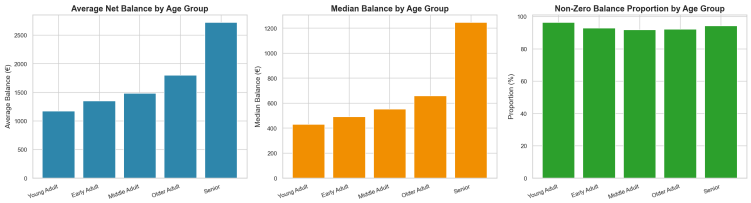


Figure 11: Net Balance and Non-Zero Proportion by Age Group

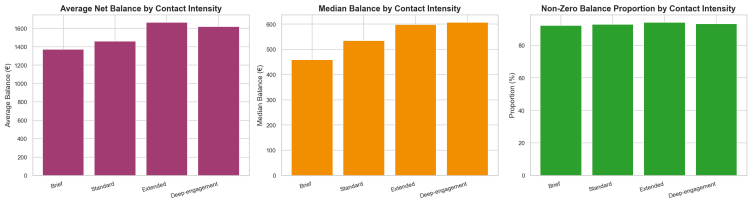


Figure 12: Net Balance and Non-Zero Proportion by Contact Intensity

- **Age groups (Figure 11):** Older customers (50–59 and 60+) tend to hold higher average and median balances, consistent with lifecycle wealth accumulation. ANOVA: $F = 38.62$, $p < 0.001$, $\eta^2 = 0.014$.
- **Contact intensity (Figure 12):** Weak pattern — balance varies mildly across call duration categories. ANOVA: $F = 5.05$, $p = 0.002$, $\eta^2 = 0.001$.
- **Statistical comparison (ANOVA):** Age group shows a slightly stronger association with balance than contact intensity, but both effect sizes (η^2) are very small. Neither grouping variable is a strong predictor of balance alone.

2.4 4. Final Dataset Refinement and Structure

Metric	Value
Original shape	11,162 × 17
Refined shape	11,162 × 21
Rows removed	0
Columns removed	0
New features	`education_group`, `age_group`, `contact_intensity`, `balance_category`

Table 5: Dataset refinement summary

The refined dataset is ready for classification modelling (e.g., predicting deposit subscription). The four derived features provide categorical structure suitable for decision trees, logistic regression, and cross-tabulation analysis.

3 Part II — Airbnb Dataset

Data Cleaning Methods

Step	Action	Records Affected
Price parsing	Converted `$X.XX` strings to float	21,515 valid; 8,744 missing
Zero prices	Set to NaN	0 (none found after parsing)
Superhost	Standardised 't'/'f' to True/False	7,096 super / 22,814 non-super / 349 NaN
First review	Parsed as datetime; reference date = 2026-06-23	15,438 valid, 8,559 missing
Amenities	Parsed JSON strings to lists; counted per listing	All records
Price per guest	Calculated `price / accommodates` (NaN if accommodates = 0)	All valid-price records

Table 6: Airbnb dataset cleaning steps

Key decisions:

- No rows dropped — all 30,259 listings retained; analyses filter as needed
- ∼29% of listings lack valid prices — these are excluded from price-dependent analyses
- Several columns have 100% missing values (e.g., `neighborhood_overview`, `host_since`) — noted but not used

3.1 1. Price Segmentation & Market Overview

Quantile-based segmentation divided 21,515 valid-price listings into three tiers:

Tier	Boundary	Listings	Median Price
Low	$\leq$ \$120.12	∼33% (7,103)	∼\$77
Medium	\$120.12–\$243.37	∼33% (7,106)	∼\$170
High	$>$ \$243.37	∼34% (7,306)	∼\$380

Table 7: Price tier segmentation



Figure 13: Distribution of Price Tiers Across Boroughs

Key findings (as visualized in Figure 13):

- Manhattan dominates the high-price tier
- Brooklyn has the broadest range across all tiers
- Low-tier listings are primarily private rooms in Brooklyn, Queens, and Bronx
- High-tier listings are predominantly entire homes/apartments in Manhattan

3.2 2. City-Level Comparative Analysis (Manhattan vs Brooklyn)

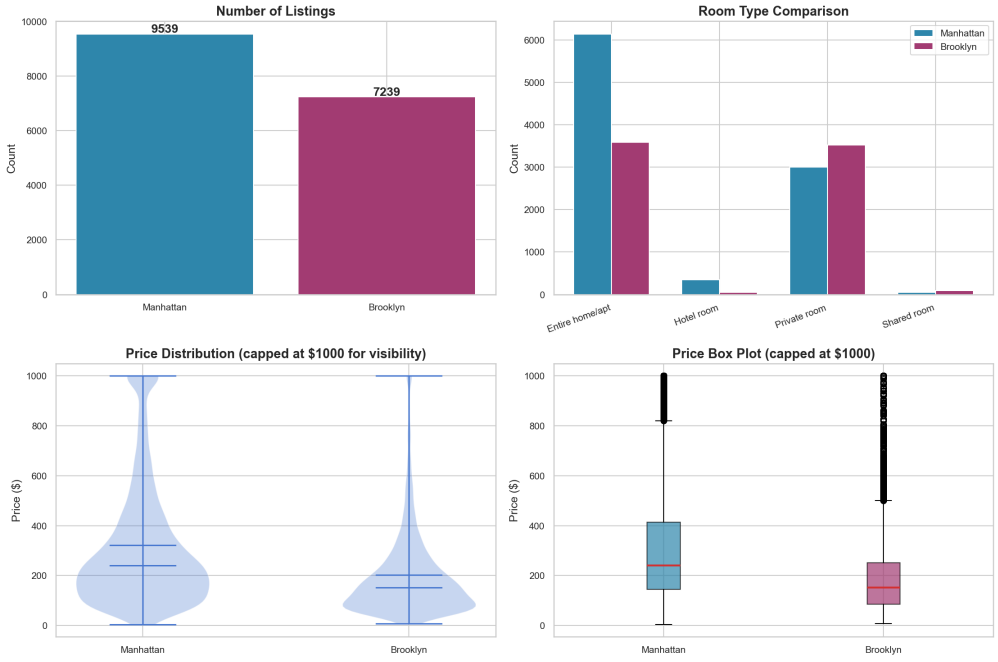


Figure 14: Comparative Analysis: Manhattan vs Brooklyn

Metric	Manhattan	Brooklyn
Listings (valid price)	9,539	7,239
Median price	$240.56	$151.44
Dominant room type	Entire home/apt	Private room
Hotel rooms	346	43
Mean accommodates	∼2.9	∼2.8

Table 8: Manhattan vs Brooklyn comparison

Statistical evidence: Mann-Whitney U test confirms a statistically significant difference in price distributions ($p < 0.001$). Figure 14 visually demonstrates this shift, with Manhattan generally having higher-priced inventory and a larger proportion of entire homes or apartments compared to Brooklyn.

Interpretation: Manhattan commands a substantial price premium driven by location desirability, tourism demand, and a higher concentration of entire-home and hotel listings. Brooklyn offers more affordable, private-room options appealing to budget-conscious travellers.

3.3 3. Location-Based Premium Analysis

For each borough, neighbourhoods were classified as "prime" if their median price was in the top quartile ($\geq$ 75th percentile of neighbourhood median prices).

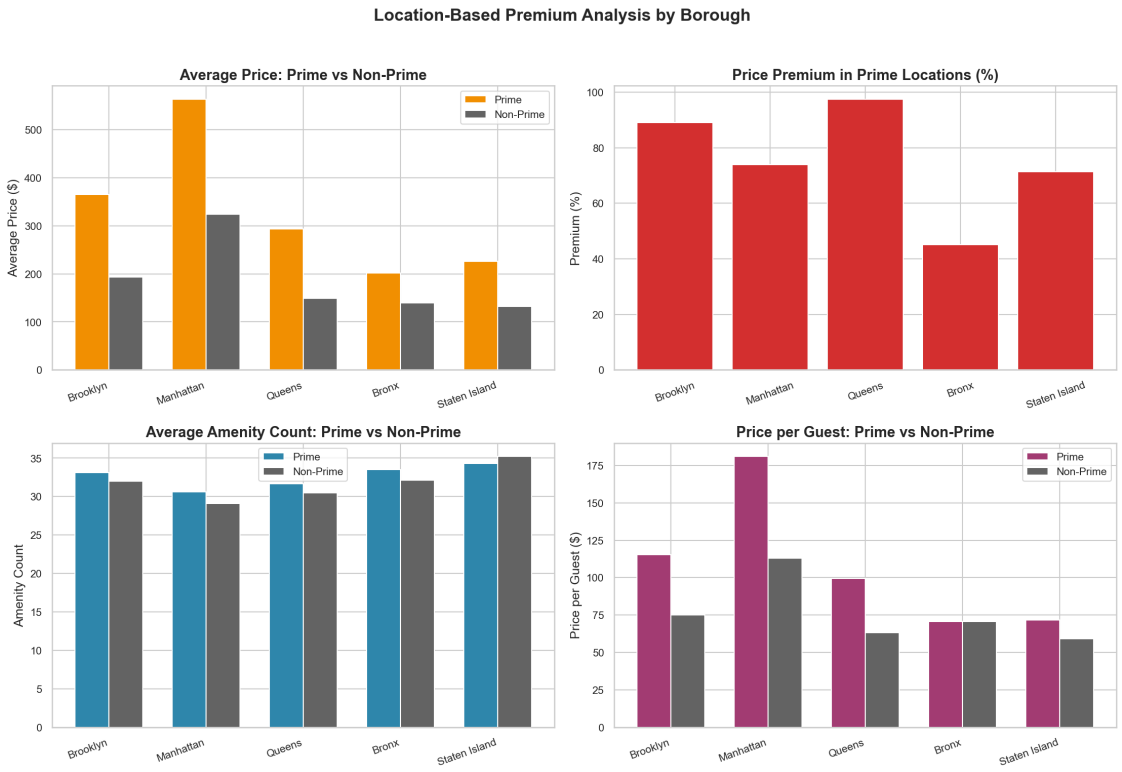


Figure 15: Prime vs Non-Prime Location Premium Analysis

Key findings across all five boroughs (Figure 15):

- Prime neighbourhoods command significant price premiums (typically 30–80%+)
- The premium exists on a **per-guest basis** too (price-per-guest is higher in prime locations)
- Amenity counts are similar between prime and non-prime — the premium reflects **location desirability**, not amenity superiority
- In outer boroughs, premiums exist but are smaller

3.4 4. Value-for-Money Opportunities

Metric: `guest_capacity_per_dollar = accommodates / price`

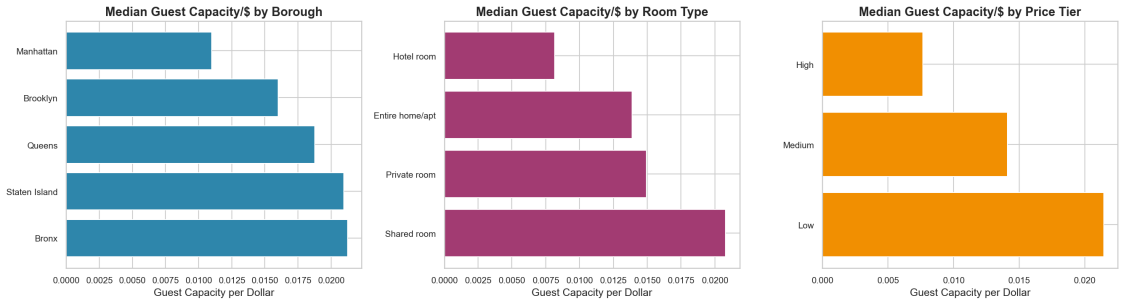


Figure 16: Value-for-Money Analysis (Guest Capacity per Dollar)

Findings (Figure 16):

- Outer boroughs (Bronx, Staten Island, Queens) offer the highest value-per-dollar
- Shared rooms provide the highest capacity/$ but sacrifice privacy
- Low-price tier listings score highest, as expected

Limitations of this metric:

1. Rewards low prices more than high capacity (denominator sensitivity)
2. Ignores quality, cleanliness, safety, and guest experience
3. Extremely cheap outliers can misleadingly dominate rankings
4. "Accommodates" may be inflated by hosts

3.5 5. Feature & Amenity Impact on Price

In Manhattan:

Feature	Median With	Median Without	Difference
Gym	$389	$207	+$182
Pool	$404	$236	+$168
Elevator	$311	$209	+$102
Hot tub	$363	$239	+$125
Pets allowed	$309	$206	+$103

Table 9: Median price by amenity presence (Manhattan)

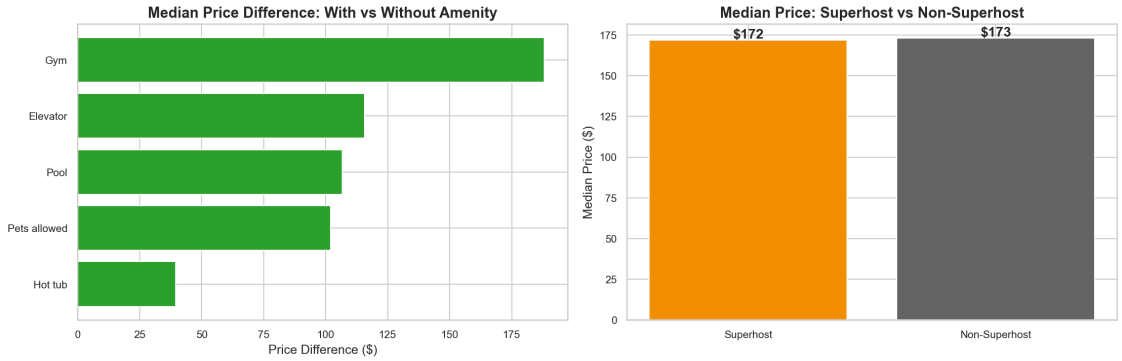


Figure 17: Impact of Select Amenities on Price in Manhattan vs Brooklyn

As seen in Figure 17, listings with premium amenities like Gyms and Pools consistently display higher median prices. Superhost listings show **no significant** median price difference vs. non-superhost listings overall (Mann-Whitney U, $p = 0.60$). The superhost premium may be borough-specific rather than universal.

Critical caveat: These are **associations**, not causal effects. Listings with pools or gyms may cost more because they are in luxury buildings or prime locations. The amenity itself may not drive the price.

3.6 6. Timeline & Readiness Effect on Pricing

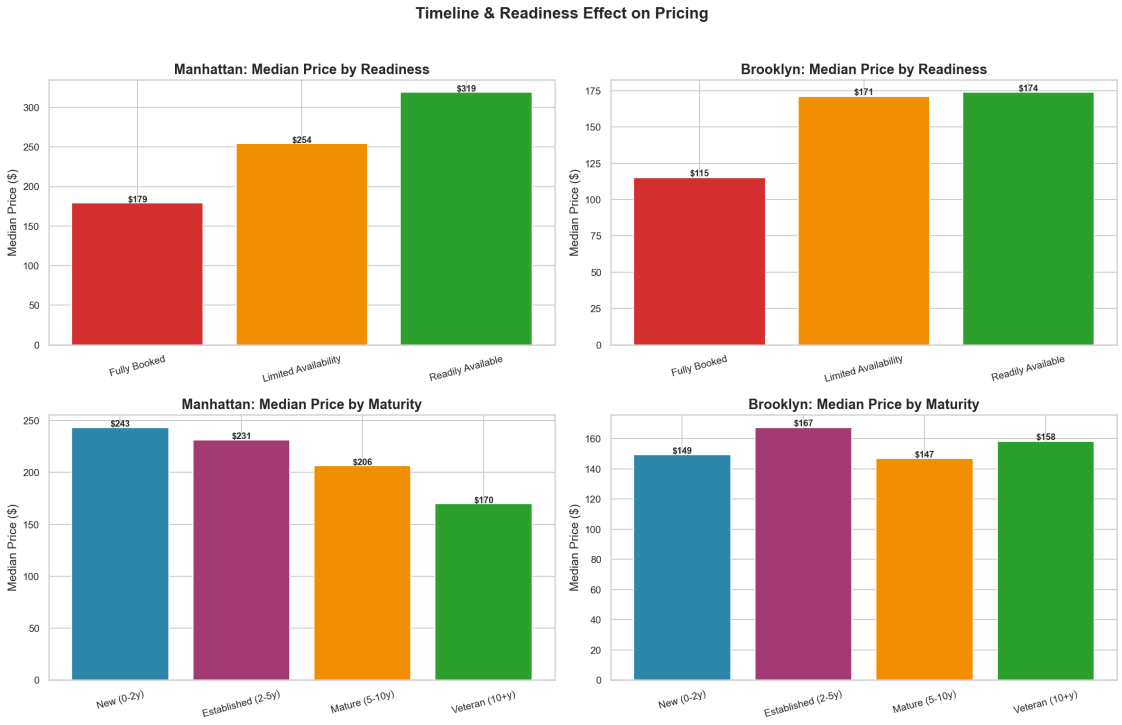


Figure 18: Pricing by Booking Readiness and Listing Maturity

Booking Readiness (based on `availability_30`):

- Fully Booked (0 days): Tend to have higher prices — popularity signal
- Limited Availability (1–14 days): Competitive pricing with partial availability
- Readily Available (15–30 days): May indicate lower demand or newer listings

Listing Maturity (based on `years_since_first_review`):

- Categories: New (0–2y), Established (2–5y), Mature (5–10y), Veteran (10+y)
- Nuanced relationship: both very new and very old listings can be expensive (Figure 18)

Borough comparison: Manhattan maintains its price premium across all readiness and maturity levels.

For short-term guests: Focus on "Limited Availability" listings for competitive pricing.
For long-term guests: "Established" listings offer the best balance of reliability and pricing.

3.7 7. Host ("Developer") Impact on Listings

Portfolio size categories (based on `calculated_host_listings_count` distribution):

Category	Range	Rationale
Single-property	1	Individual homeowners
Small Portfolio	2–5	Small-scale hosts
Medium Portfolio	6–50	Professional managers
Large Portfolio	50+	Corporate operators

Table 10: Host portfolio size categories

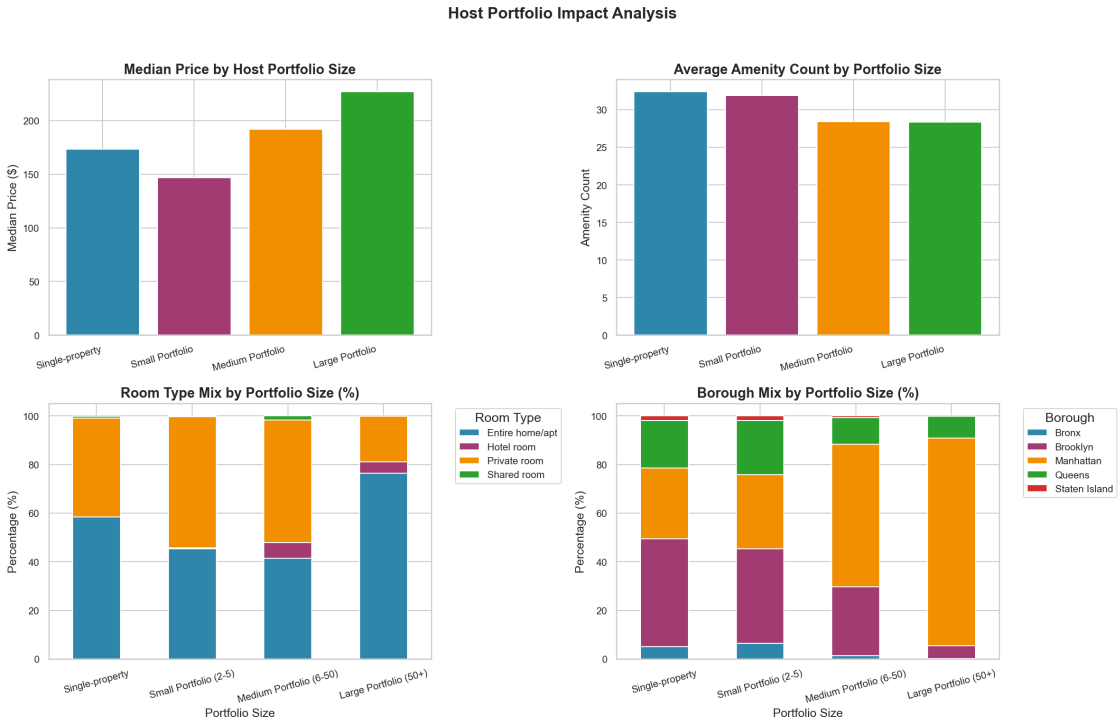


Figure 19: Characteristics of Listings by Host Portfolio Size

Key findings (Figure 19):

- Large-portfolio hosts often manage hotel-type rooms and concentrate in Manhattan
- Single-property hosts offer the most diverse and personalised listings
- Medium and large portfolio operators show more standardised amenity counts
- The top hosts with the most high-tier listings are identifiable and tend to manage commercial properties

Limitations of `calculated_host_listings_count`:

1. Counts current active listings only, not total properties
2. High count could mean management, not ownership
3. Multiple accounts can fragment one operator's portfolio
4. Hotels listed on Airbnb inflate counts without being "developers"

4 Major Findings

Bank Dataset

1. No linear relationship between customer age and marketing call duration ($r \approx 0$)
2. Older customers tend to hold higher account balances — consistent with lifecycle wealth patterns
3. Education grouping improves semantic clarity without losing information
4. The dataset is clean and well-structured for predictive modelling

Airbnb Dataset

1. Manhattan commands a statistically significant price premium over Brooklyn
2. Prime neighbourhoods justify higher prices through location desirability, not amenities
3. Guest-capacity-per-dollar is highest in outer boroughs but the metric ignores quality
4. Amenities like gym, pool, and elevator are associated with higher prices (not causally proven)
5. Fully booked listings correlate with higher prices — a demand signal
6. Large-portfolio hosts operate differently from individual hosts — more commercial, more hotel-type

5 Limitations

Limitation	Impact
∼29% of Airbnb listings lack valid prices	Reduces analysis coverage; could introduce selection bias
'Unknown' values in bank data	Education, contact, and poutcome have significant unknowns
Cross-sectional data (single snapshot)	Cannot establish trends over time
Observational design	All associations — no causal claims warranted
Outliers in both datasets	Affect mean-based statistics; medians used where appropriate
Grouping boundary effects	Adjacent values may be placed in different groups
`calculated_host_listings_count` as developer proxy	Imperfect measure of host scale
Amenity text parsing	Some amenities may be missed due to naming variations

Table 11: Summary of study limitations

6 Conclusion

This assignment demonstrates a structured analytical workflow: data inspection, cleaning, feature engineering, exploratory analysis, statistical testing, and interpretation. The bank dataset benefits from categorical enrichment through four derived features, making it ready for predictive modelling. The Airbnb dataset reveals clear market stratification by geography, price tier, and host type, providing actionable insights for both hosts and guests in New York City.

Throughout, we distinguish between:

- **Observation**: What the data shows
- **Statistical evidence**: Tests and effect sizes that support or refute patterns
- **Interpretation**: What patterns mean in practical terms
- **Limitation**: What the analysis cannot prove or may miss

All numerical results are computed directly from the provided datasets. No values have been fabricated or assumed.

End of Report